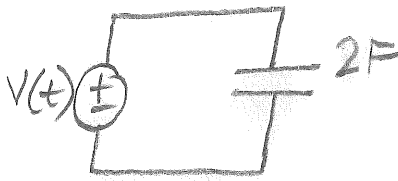


EBN111 Klastoets/Class Test#3: 24 Mei/May 2013

- Q1 Bereken $i(t)$ as $v(t) = (2 - 4e^{-2t})$ V.
Calculate $i(t)$ if $v(t) = (2 - 4e^{-2t})$ V.

[1]



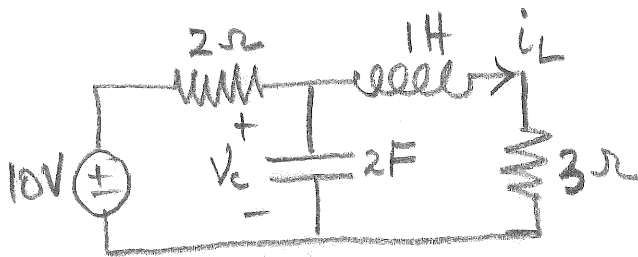
$$i = C \frac{dv}{dt} = 2 \times \frac{d}{dt} (2 - 4e^{-2t})$$

$$= -2 \times 4 \times e^{-2t} \times (-2)$$

$$= 16 e^{-2t} \text{ A} \checkmark$$

- Q2 Bereken i_L en v_C .
Determine i_L and v_C .

[2]

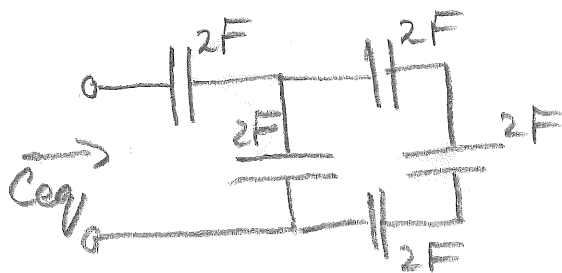


$$i_L = \frac{10}{5} = 2 \text{ A} \checkmark$$

$$v_C = \frac{3}{5} \times 10 = 6 \text{ V} \checkmark$$

- Q3 Bereken C_{eq} tov terminale a-b.
Determine C_{eq} wrt terminals a-b.

[2]



$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

$$\frac{2}{3} + 2 = 2\frac{2}{3} = \frac{8}{3}$$

$$\frac{3}{8} + \frac{1}{2} = \frac{7}{8} \Rightarrow C_{eq} = \frac{8}{7}$$

$$= 1.143 \text{ F} \checkmark$$

- Q4 Skryf $\bar{V} = 15 \angle -20^\circ$ V as 'n cos-funksie, en skryf $i(t) = 3 \sin(2t - 20^\circ)$ A as 'n fasor (in polêre form).
Write $\bar{V} = 15 \angle -20^\circ$ V as a cos-function, and write $i(t) = 3 \sin(2t - 20^\circ)$ A as a phasor (in polar form).

[2]

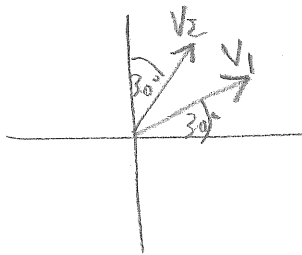
$$v(t) = 15 \cos(\omega t - 20^\circ) \text{ V} \checkmark$$

$$I = 3 \angle -110^\circ \text{ A} \checkmark$$

- Q5 Watter een van die volgende sinusiede loop voor ($v_1(t)$ of $v_2(t)$), en met hoeveel grade.
Which of the following sinusoids leads ($v_1(t)$ of $v_2(t)$), and by how many degrees.

[2]

$$v_1(t) = 20 \cos(2t + 30^\circ) \text{ V}, v_2(t) = -10 \sin(2t - 30^\circ) \text{ V}$$



$v_2(t)$ loop voor met 30°

- Q6 Skryf $\bar{Z}_1 = 10 \angle -30^\circ \Omega$ in reghoekige vorm en $\bar{Z}_2 = -5 + j10 \Omega$ in polêre vorm.
Write $\bar{Z}_1 = 10 \angle -30^\circ \Omega$ in rectangular form and $\bar{Z}_2 = -5 + j10 \Omega$ in polar form.

[2]

$$\bar{Z}_1 = 8.66 - j5 \checkmark$$

$$\bar{Z}_2 = 11.18 \angle 116.6^\circ \checkmark$$

- Q7 Vereenvoudig en skryf die finale antwoord in polêre vorm.
Simplify and write final answer in polar form.

[2]

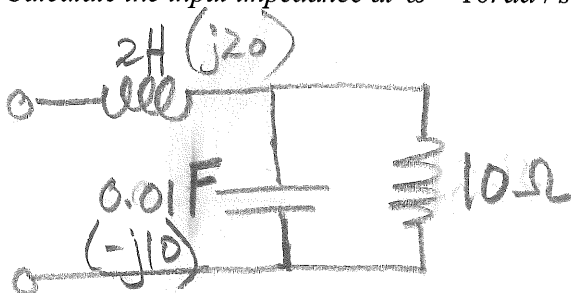
$$\bar{Z}_3 = \frac{2 \angle 75^\circ + 3 - j4}{2 + j3} = \frac{0.518 + j1.93 + 3 - j4}{2 + j3}$$

$$= \frac{3.518 - j2.07}{2 + j3} = \frac{4.08 \angle -30.5^\circ}{3.61 \angle 56.3^\circ}$$

$$= 1.13 \angle -86.8^\circ$$

- Q8 Bereken die intree-impedansie by $\omega = 10 \text{ rad/s}$.
Calculate the input impedance at $\omega = 10 \text{ rad/s}$.

[2]



$$10 \parallel -j10$$

$$= \frac{10 \times (-j10)}{10 - j10} = \frac{100 \angle -90^\circ}{14.14 \angle -45^\circ}$$

$$= 7.07 \angle -45^\circ = 5 - j5$$

$$\bar{Z}_{in} = 5 - j5 + j20 = 5 + j15 \Omega$$

$$r = \sqrt{x^2 + y^2}$$

$$15.81 \angle 71.57^\circ$$